

Pre-Registered Blind Comparison Protocol for the Yang-Mills Mass Gap Construction

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Abstract

We present the formal mathematical framework, pre-registered predictions, and automated checking methodology for evaluating the newly arrived 600-page constructive proof of the Yang-Mills mass gap. Following the blind-comparison protocol established on July 4, 2026, we categorize our predictions into clean (scoreable) and contaminated (non-scoreable) sets, mapping them to the expected lemma structure of the construction. We detail the design of an automated, self-healing sliding-window density parser used to scan the construction and generate verifiable matching ledgers.

1 Introduction

The Yang-Mills mass gap problem stands as one of the Millennium Prize Problems in mathematical physics, requiring the proof of the existence of a quantum Yang-Mills theory on $\mathbb{R}^4$ with a gauge group G (compact, simple Lie group) and a non-zero physical mass gap $\Delta > 0$.

On July 4, 2026, we pre-registered a set of six clean predictions (C1–C6) and two contaminated predictions (X1–X2) in a sealed file (SHA-256 hash: `b449243813...`). With the arrival of the counterparty’s 600-page construction, we release the sealed document and present the formal protocol for the verification of these predictions.

2 Formal Pre-Registered Predictions

2.1 Clean Predictions (Scoreable)

Clean predictions are derived strictly from our program’s primitives alone and are eligible for scoring under the comparison protocol.

C1 Kinematic Nature of Finite Resolution (Casimir Gap): At any finite resolution, the mass gap is kinematic. The compactness of the gauge group makes the electric spectrum discrete:

$$\Delta_{\text{kin}} \approx \frac{3}{2}g^2 + \mathcal{O}(g^4) \quad (1)$$

The difficulty of the proof resides not in proving the existence of a gap at finite cutoff, but in the refinement transport.

C2 Survival under Refinement (Refinement Transport): The proof is structurally a survival theorem showing that the physical mass gap does not close along the coherent refinement path (as volume $V \rightarrow \infty$ and lattice spacing $a \rightarrow 0$ at fixed physical scale):

$$\lim_{a \rightarrow 0} \Delta(a) \geq \Delta_0 > 0 \quad (2)$$

C3 Gauge-Invariant Algebra Projection: The massless-gluon obstruction is a projection-class artifact of the gauge-variant description (e.g., the perturbative propagator). The proof bypasses this by working entirely inside gauge-invariant observable algebras, excluding the massless artifact by construction.

C4 One-Channel Law (0^{++} Plaquette): The quantitative bound lives entirely in the invariant 0^{++} plaquette-plaquette correlation channel (transfer-matrix spectrum). The hard estimate is a uniform bound on the correlation length:

$$\langle \Phi(x)\Phi(0) \rangle_c \leq C e^{-|x|\Delta} \quad (3)$$

C5 Exchange of Limits (Cutoff Uniformity): The non-commuting limits of weak coupling and infinite volume are resolved via uniform, frame-independent bounds carrying no cutoff dependence.

C6 Constructive Proof Form: The result is constructive, exhibiting the vacuum state and the gap via reflection positivity and a uniform correlation-length estimate across a scale sequence, rather than a compactness or existence-only argument.

2.2 Contaminated Predictions (Non-Scoreable)

Contaminated predictions are informed by counterparty clues (e.g., Wilson-type algebras, admissibility, refinement transport) and are excluded from the landed-same-spot score.

X1 Wilson-Type Algebras: Refinement transport between nested Wilson-type algebras $\mathcal{A}_n^\epsilon(\Omega)$ inside gauge-invariant $\mathcal{O}_n^{\text{gi}}(\Omega)$ under admissibility ϵ -control.

X2 Observable Algebra Pivot: Utilizing Algebraic Quantum Field Theory (AQFT) or observable algebra representations.

3 Automated Comparison Checker Methodology

To eliminate human evaluation bias, we constructed the comparison checker tool, `ym_comparison_checker.py`. The checker operates via the following steps:

1. **Input Extraction:** Recursively parses targeted LaTeX, text, or PDF files. PDF files are parsed via a self-healing import of the `pypdf` library, auto-installed on demand.
2. **Sliding-Window Density Scan:** The parsed text is tokenized into sliding windows of $W = 150$ words.
3. **Keyword Co-occurrence Proximity:** For each prediction P , the tool evaluates the presence of must-have and should-have keyword clusters. The window match score is defined as:

$$S(P) = 0.6 \left(\frac{|W \cap \text{Must}(P)|}{|\text{Must}(P)|} \right) + 0.4 \left(\frac{|W \cap \text{Should}(P)|}{|\text{Should}(P)|} \right) \quad (4)$$

4. **Verdict Classification:** Scores $S(P) \geq 0.7$ map to **MATCH**; $0.25 \leq S(P) < 0.7$ map to **PARTIAL**; others map to **MISS**.

4 Verification Protocol

Each prediction is mapped to the newly arrived lemma structure. In the event of a **MISS**, the result is treated as first-class, localizing the divergence between our theoretical framework and the constructive proof. This divergence forms the basis for joint-paper development.